

Backend Engine Specification

The services that must be built behind the Public website, Borrower self-service, MFI Member Portal and Government Portal — their scope, which portals use them, and what they integrate with.

25

backend engines

5

portals / portal areas

14

external integrations

16 / 7 / 2

Pilot / Production / Option

Document CIC Myanmar Platform — Backend Engines, scope & integrations

Version 1.0 draft for internal review

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Based on CIC Myanmar Platform — E2E Requirements Specification v1.0; working prototype (cic-platform.vercel.app)

1. Summary

The front-end prototype runs every portal on data kept in the browser. For production, that logic moves into 25 backend engines behind one API gateway. Each engine owns its data, exposes versioned APIs, and writes every sensitive action to the audit store. Portals never talk to a database directly.

One core, many portals. Registry, inquiry, disputes and billing are built once and reused by every portal.

Maker-checker everywhere. Licence changes, corrections, rule activation, publishing, role changes and large loans go through one approval engine.

Bilingual. Content and UI strings are served in English and Myanmar (Unicode) from the CMS.

Security enforced on the server. Tenant, role, CRUD permission and data scope are checked by the permission decision point on every call.

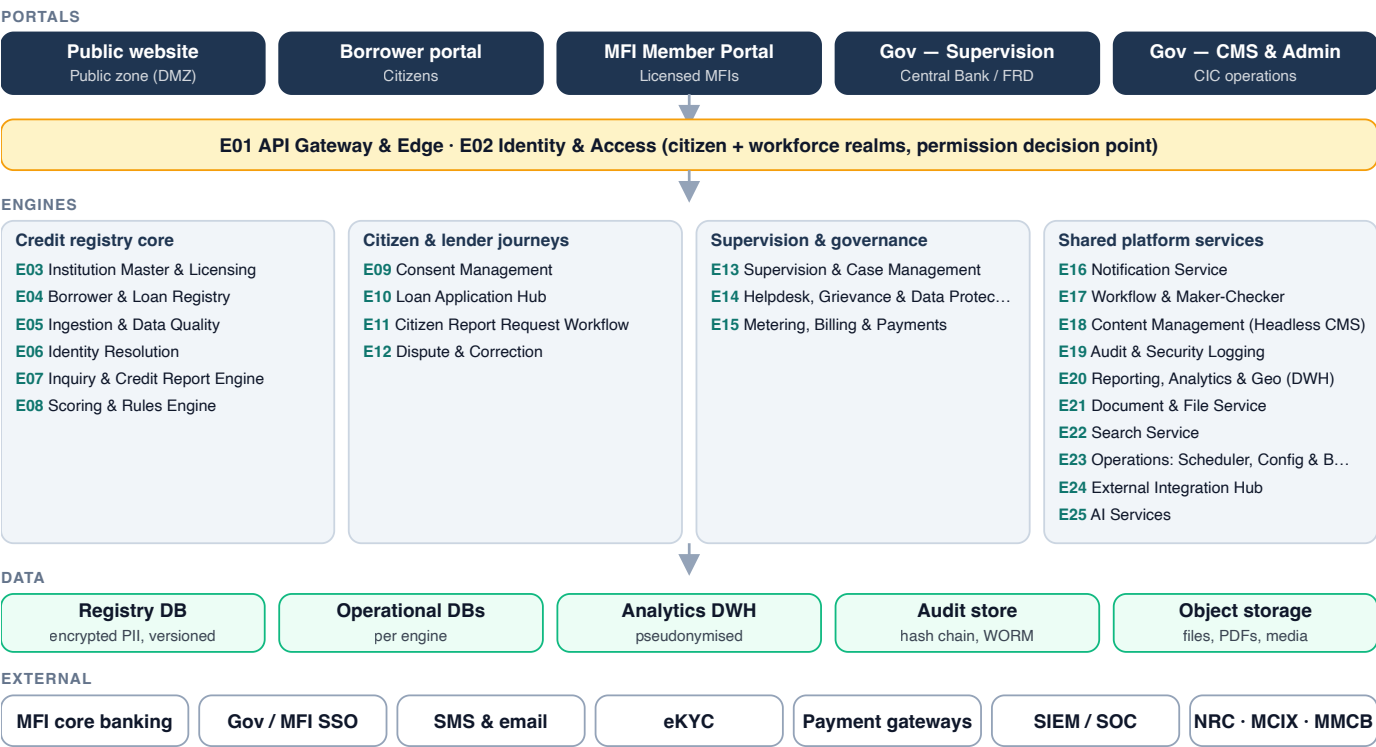
Evidence by design. Hash-chained audit, consent references, rule versions and data dates travel with every report.

Integrations as options. NRC, MCIX, MMCB and alternative data sit behind an integration hub with their own agreements.

PHASE LEGEND

Pilot needed before the 5-MFI pilot · **Production** needed before scaling to 30 MFIs · **Option** after the foundation is stable, subject to agreements.

2. Architecture overview



3. Engine catalogue

ID	Engine	Group	Phase	Purpose
E01	API Gateway & Edge	Edge & security	Pilot	Single entry point for every portal and machine client; nothing talks to a database directly.
E02	Identity & Access Management (IAM)	Edge & security	Pilot	Two identity realms — citizens and workforce — plus the permission decision point used by every service.
E03	Institution Master & Licensing	Credit registry core	Pilot	One record per MFI that drives the public directory, supervision, billing and ingestion validation.
E04	Borrower & Loan Registry	Credit registry core	Pilot	The golden record of borrowers, loans, guarantees and repayment history — versioned and auditable.
E05	Ingestion & Data Quality	Credit registry core	Pilot	Takes monthly/daily submissions from MFIs, validates them and loads clean data into the registry.
E06	Identity Resolution	Credit registry core	Pilot	Decides which borrower a record belongs to — and never auto-merges look-alikes.
E07	Inquiry & Credit Report Engine	Credit registry core	Pilot	Purpose-bound, consented inquiries that produce Basic/Full reports, PDFs and verifiable snapshots.
E08	Scoring & Rules Engine	Credit registry core	Pilot	Versioned rule sets for credit grade, early-warning and over-indebtedness — the same result everywhere.
E09	Consent Management	Citizen & lender journeys	Pilot	Proof that every inquiry was permitted, and a place for citizens to see and revoke consents.
E10	Loan Application Hub	Citizen & lender journeys	Production	Citizens apply online to a licensed MFI; the MFI checks, decides and disburses; the loan lands in the registry.
E11	Citizen Report Request Workflow	Citizen & lender journeys	Production	A citizen asks for their own report; CIC validates the data, an officer approves and the report is issued.
E12	Dispute & Correction	Citizen & lender journeys	Pilot	Borrower disputes flow to the reporting MFI, then to CIC for approved corrections — with SLA clocks.
E13	Supervision & Case Management	Supervision & governance	Production	Turns alerts and complaints into cases with owners, deadlines, decisions and actions.
E14	Helpdesk, Grievance & Data Protection	Supervision & governance	Production	Service desk for citizens and the DPO toolset for privacy obligations.
E15	Metering, Billing & Payments	Supervision & governance	Production	Charges for reports and subscriptions, keeps the ledger and reconciles every payment.
E16	Notification Service	Shared platform services	Production	Every SMS, email, in-app message and webhook, from EN/MM templates with a delivery log.
E17	Workflow & Maker-Checker	Shared platform services	Pilot	One approval engine for every sensitive change so the rules are enforced the same way everywhere.
E18	Content Management (Headless CMS)	Shared platform services	Pilot	All public and portal content, bilingual, with workflow and classification enforced by the API.
E19	Audit & Security Logging	Shared platform services	Pilot	Tamper-evident record of who did what, to which record, for what purpose.
E20	Reporting, Analytics & Geo (DWH)	Shared platform services	Pilot	One source of numbers — KPI dictionary, dashboards, statutory reports, open data and maps.
E21	Document & File Service	Shared platform services	Pilot	Storage and processing for every uploaded or generated file.

ID	Engine	Group	Phase	Purpose
E22	Search Service	Shared platform services	Production	Site-wide search for the public website and quick finders in the portals.
E23	Operations: Scheduler, Config & Backup	Shared platform services	Pilot	Runs the platform: jobs, configuration, feature flags, backups and disaster recovery.
E24	External Integration Hub	Shared platform services	Option	Adapters to national and partner systems, each behind its own agreement.
E25	AI Services	Shared platform services	Option	Assistive AI kept separate from decisions that affect citizens.

4. Which portals use which engine

Engine	Public website	Borrower self-service	MFI Member Portal	Government — Supervision	Government — CMS & Administration
E01 API Gateway & Edge	●	●	●	●	●
E02 Identity & Access Management (IAM)	●	●	●	●	●
E03 Institution Master & Licensing	●		●	●	●
E04 Borrower & Loan Registry		●	●	●	●
E05 Ingestion & Data Quality			●	●	●
E06 Identity Resolution			●		●
E07 Inquiry & Credit Report Engine	●	●	●		●
E08 Scoring & Rules Engine		●	●	●	●
E09 Consent Management		●	●		
E10 Loan Application Hub		●	●	●	
E11 Citizen Report Request Workflow		●		●	●
E12 Dispute & Correction		●	●	●	●
E13 Supervision & Case Management			●	●	
E14 Helpdesk, Grievance & Data Protection	●	●		●	●
E15 Metering, Billing & Payments		●	●		●
E16 Notification Service		●	●	●	●
E17 Workflow & Maker-Checker			●	●	●
E18 Content Management (Headless CMS)	●	●	●	●	●
E19 Audit & Security Logging		●	●	●	●
E20 Reporting, Analytics & Geo (DWH)	●		●	●	●
E21 Document & File Service	●	●	●	●	●
E22 Search Service	●				●
E23 Operations: Scheduler, Config & Backup					●
E24 External Integration Hub		●	●	●	
E25 AI Services	●			●	

Public website and CMS-published content are served from the public zone and never hold personal data; all other portals sit behind the gateway in the private zone.

5. Engine details — scope and integrations

E01 API Gateway & Edge

Pilot Architecture §3, SEC-06, §11.1

Single entry point for every portal and machine client; nothing talks to a database directly.

SCOPE

- Routing of public zone (DMZ) and private zone APIs; versioned REST (/v1)
- WAF, rate limiting, bot protection, DDoS shield on public endpoints
- mTLS + OAuth2 client-credentials for MFI system-to-system APIs
- Token validation, tenant ID injected from token (never from request body)
- Request/correlation IDs propagated to Audit & Logging

USED BY PORTALS

Public website Borrower self-service
MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E02 Identity & Access Management (IAM)
E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

CDN / DNS provider
MFI core banking systems (API clients)
National CERT / SOC (threat feeds)

KEY APIS

All /v1/* routes
Public API (directory, notices, statistics)
Report verification API

DATA OWNED

Gateway config, API client registry, rate-limit counters

Two identity realms — citizens and workforce — plus the permission decision point used by every service.

SCOPE

- Citizen realm: registration with NRC + phone + own password, identity verification (selfie eKYC / SMS to phone on record at an MFI / branch activation code), SMS OTP on every sign-in, lockout after 5 failures
- Workforce realm: CIC, Central Bank and MFI staff; SSO (OIDC/SAML) + TOTP/FIDO2; IP allow-lists; session policy 10 min idle / 8 h absolute
- Policy decision point: role catalogue, CRUD + Approve + Export matrix per feature, data scope (all / regional / own institution / aggregates / masked PII)
- Role builder and user assignment with maker-checker; MFI role templates assigned by MFI administrators
- Dormant-account auto-disable (90 days), access recertification campaigns, break-glass accounts

USED BY PORTALS

Public website Borrower self-service
MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E17 Workflow & Maker-Checker
E18 Content Management (Headless CMS)
E19 Audit & Security Logging
E20 Reporting, Analytics & Geo (DWH)

INTEGRATES WITH — EXTERNAL

Government SSO / directory (CBM, CIC)
MFI identity providers (SAML/OIDC)
SMS gateway (OTP) eKYC / face-match provider
National NRC registry (option)

KEY APIS

/v1/auth/citizen/*
/v1/auth/workforce/*
/v1/iam/roles
/v1/iam/users
/v1/iam/decide (PDP)

DATA OWNED

Citizen accounts, staff users, roles & permission matrix, sessions, MFA factors

E03 Institution Master & Licensing

Pilot C2, GOV-03, ADM-05, PUB-05

One record per MFI that drives the public directory, supervision, billing and ingestion validation.

SCOPE

- Institution register: licence number, tier, capital, branches, contacts, products, geo location
- Licence status lifecycle (Licensed / Under review / Suspended / Revoked) with history; changes by Licensing Officer, approved by Director
- Publish flag feeding the public MFI directory
- Reference data: regions, townships, product types, purpose codes, dispute reasons, currencies, holidays

USED BY PORTALS

Public website MFI Member Portal

Government — Supervision

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E17 Workflow & Maker-Checker

E19 Audit & Security Logging

E15 Metering, Billing & Payments

E13 Supervision & Case Management

INTEGRATES WITH — EXTERNAL

FRD licensing system (option)

Myanmar township/region reference data

KEY APIS

/v1/institutions
/v1/institutions/{id}/licence
/v1/reference/*

DATA OWNED

Institutions, licence history, branches, reference tables

E04 Borrower & Loan Registry

Pilot C3

The golden record of borrowers, loans, guarantees and repayment history — versioned and auditable.

SCOPE

- Borrower master (resolved identity, NRC incl. previous NRC, names MM/EN, demographics, address)
- Loans, guarantors, group membership, 24-month repayment history, classification, write-offs, closure
- Record versioning: every correction creates a new version with lineage to the dispute or batch
- Loans disbursed through online applications posted immediately and reconciled with the next monthly batch
- Retention and purge by data class

USED BY PORTALS

Borrower self-service MFI Member Portal

Government — Supervision

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E05 Ingestion & Data Quality E06 Identity Resolution

E07 Inquiry & Credit Report Engine

E09 Consent Management

E11 Citizen Report Request Workflow

E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

None

KEY APIS

Internal only (via Inquiry, Ingestion, Disputes, Analytics)

DATA OWNED

Registry DB (encrypted fields: NRC, phone, address), record versions

E05 Ingestion & Data Quality

Pilot C4, MFI-07..12, ADM-06, AC01/AC02

Takes monthly/daily submissions from MFIs, validates them and loads clean data into the registry.

SCOPE

- Intake: CSV / XLSX / XML upload, SFTP and Submission API (idempotency key = batch ID)
- Schema versions (v3.x), downloadable templates, DQ rules (errors / warnings), row-level validation report
- Maker-checker sign-off with attestation; bulk sign-off for checkers
- Reconciliation against control totals; reject queue, re-process, resubmission without duplicates
- Submission calendar, cut-offs, late flags; DQ score per batch and per MFI

USED BY PORTALS

MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E06 Identity Resolution E04 Borrower & Loan Registry
E17 Workflow & Maker-Checker
E18 Content Management (Headless CMS)
E19 Audit & Security Logging
E21 Document & File Service

INTEGRATES WITH — EXTERNAL

MFI core banking / loan systems (API, SFTP)
Object storage (raw files)

KEY APIS

POST /v1/batches
GET /v1/batches/{id}/validation
/v1/batches/{id}/signoff
/v1/schemas

DATA OWNED

Batches, rows staging, validation results, reconciliation, DQ rules

E06 Identity Resolution

Pilot C5, ADM-07, AC03, LOC-02/03

Decides which borrower a record belongs to — and never auto-merges look-alikes.

SCOPE

- NRC parsing and normalisation (Myanmar/Latin digits, township codes, citizenship type)
- Name normalisation (honorifics U/Daw/Ko/Ma, Unicode, Zawgyi detection)
- Matching: single match / multiple candidates / no-hit; similarity scores
- Merge / not-same / split review queue with maker-checker and full lineage

USED BY PORTALS

MFI Member Portal
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry
E17 Workflow & Maker-Checker
E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

National NRC registry (option)

KEY APIS

Internal: /resolve, /candidates; Admin:
/v1/identity/pairs

DATA OWNED

Match candidates, decisions, lineage graph

E07 Inquiry & Credit Report Engine

Pilot C6, MFI-02..06, PUB-09, AC05

Purpose-bound, consented inquiries that produce Basic/Full reports, PDFs and verifiable snapshots.

SCOPE

- Search by NRC or CIC borrower ID; purpose code and consent reference mandatory
- Report assembly: Basic (summary, exposure, delinquency flag, grade) and Full (24-month grid, guarantees, inquiries, dispute flags, reason codes)
- Entitlement check: report unlocked per institution per borrower for 30 days (pay-per-report) or covered by subscription
- Citizen personal report snapshots (issued after CIC approval, valid 30 days)
- PDF rendering with QR, watermark, rule version and data date; public verification (valid / revoked / expired / not found)
- Inquiry log that feeds "Who viewed my report"

USED BY PORTALS

Public website Borrower self-service

MFI Member Portal

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry

E08 Scoring & Rules Engine E09 Consent Management

E15 Metering, Billing & Payments

E19 Audit & Security Logging

E21 Document & File Service

INTEGRATES WITH — EXTERNAL

MFI loan origination systems (Inquiry API)

KEY APIS

POST /v1/inquiries

GET /v1/reports/{id}

GET /v1/reports/{id}/pdf

GET /v1/verify?id=&code=

DATA OWNED

Inquiry log, report snapshots, verification codes

E08 Scoring & Rules Engine

Pilot C6, ADM-08, GOV-08/09/19, S4

Versioned rule sets for credit grade, early-warning and over-indebtedness — the same result everywhere.

SCOPE

- Credit grade A–E with points and reason codes (no-hit returns no grade, never a low score)
- EWS rules library (PAR30, late submissions, concentration...) generating alerts
- Over-indebtedness rules (≥ 3 active loans, debt-to-income thresholds) by township
- Rule-set lifecycle: draft → test on sample → approve → activate with effective date → rollback
- Policy impact simulation on historical data; statistical score model as a later option

USED BY PORTALS

Borrower self-service MFI Member Portal

Government — Supervision

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry

E07 Inquiry & Credit Report Engine

E13 Supervision & Case Management

E17 Workflow & Maker-Checker

E11 Citizen Report Request Workflow

INTEGRATES WITH — EXTERNAL

None

KEY APIS

Internal: /score, /ews/evaluate; Admin: /v1/rulesets

DATA OWNED

Rule sets and versions, simulation results

E09 Consent Management

Pilot G16, MFI-03, BOR-07

Proof that every inquiry was permitted, and a place for citizens to see and revoke consents.

SCOPE

- Digital consent from the borrower portal (one inquiry, one application, 30-day validity) and branch-captured consent with document reference
- Consent verification at inquiry time (not expired, not used, right institution and purpose)
- Consent log per citizen; revocation where law allows (e.g. alternative data)

USED BY PORTALS

Borrower self-service MFI Member Portal

INTEGRATES WITH — INTERNAL

E07 Inquiry & Credit Report Engine
E10 Loan Application Hub E19 Audit & Security Logging
E21 Document & File Service

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/consents
/v1/consents/{ref}/verify

DATA OWNED

Consents, evidence documents, usage

E10 Loan Application Hub

Production Citizen journey 3, MFI applications

Citizens apply online to a licensed MFI; the MFI checks, decides and disburses; the loan lands in the registry.

SCOPE

- Application submission (product, amount, tenor, income, purpose) with affordability and over-indebtedness warnings
- Routing to the chosen MFI inbox; branch-captured applications
- Credit check via Inquiry (purchase Basic/Full or subscription), decision (approve / decline with reason), second approval above threshold
- Disbursement → registry posting; status tracking and notifications to the applicant; withdrawal

USED BY PORTALS

Borrower self-service MFI Member Portal
Government — Supervision

INTEGRATES WITH — INTERNAL

E07 Inquiry & Credit Report Engine
E09 Consent Management
E04 Borrower & Loan Registry
E15 Metering, Billing & Payments
E16 Notification Service
E17 Workflow & Maker-Checker
E13 Supervision & Case Management

INTEGRATES WITH — EXTERNAL

MFI loan origination systems (webhooks)

KEY APIS

/v1/applications
/v1/applications/{id}/decision
/v1/applications/{id}/disburse

DATA OWNED

Applications, decisions, status history

E11 Citizen Report Request Workflow

Production Citizen journey 2, BOR-04/05

A citizen asks for their own report; CIC validates the data, an officer approves and the report is issued.

SCOPE

- Request (purpose, notification channel, free quota 1 per 12 months, fee thereafter, free re-issue after a corrected dispute)
- Automated validation: identity match, data freshness per lender, open disputes, pending corrections, quota
- Officer review queue with SLA (1 working day), approve & issue snapshot or reject with reason
- Notification on issue/rejection; expiry after 30 days

USED BY PORTALS

Borrower self-service
Government — CMS & Administration
Government — Supervision

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry
E07 Inquiry & Credit Report Engine
E08 Scoring & Rules Engine
E15 Metering, Billing & Payments
E16 Notification Service E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/report-requests
/v1/report-requests/{id}/decision

DATA OWNED

Requests, automated checks, decisions

E12 Dispute & Correction

Pilot C7, BOR-08/09, MFI-13, ADM-09, GOV-11, AC06

Borrower disputes flow to the reporting MFI, then to CIC for approved corrections — with SLA clocks.

SCOPE

- Dispute filing (record, reason code, evidence upload with virus scan) by citizen or helpdesk
- MFI response within 10 working days; correction proposal (field, old, new)
- CIC approval creating a new registry version; rejection back to MFI
- SLA timers, escalation to Central Bank consumer protection, dispute flags on reports

USED BY PORTALS

Borrower self-service MFI Member Portal
Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry
E07 Inquiry & Credit Report Engine
E15 Metering, Billing & Payments
E16 Notification Service
E17 Workflow & Maker-Checker
E21 Document & File Service

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/disputes
/v1/disputes/{id}/response
/v1/disputes/{id}/correction

DATA OWNED

Disputes, evidence, SLA events, corrections

E13 Supervision & Case Management

Production GOV-04..12

Turns alerts and complaints into cases with owners, deadlines, decisions and actions.

SCOPE

- EWS alert queue (from Scoring & Rules), triage, dismissal with reason
- Supervisory cases: investigate, information requests to MFIs, decision (Director approval), action (warning, fine, suspension)
- Complaint register fed by the public grievance form; linkage to MFIs
- Borrower-level drill-down only with logged justification; regional scope for regional supervisors

USED BY PORTALS

Government — Supervision MFI Member Portal

INTEGRATES WITH — INTERNAL

E08 Scoring & Rules Engine
E03 Institution Master & Licensing
E14 Helpdesk, Grievance & Data Protection
E16 Notification Service
E17 Workflow & Maker-Checker
E19 Audit & Security Logging
E21 Document & File Service

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/alerts
/v1/cases
/v1/info-requests
/v1/complaints

DATA OWNED

Alerts, cases, correspondence, decisions

E14 Helpdesk, Grievance & Data Protection

Production ADM-09/14, PUB-12, BOR-11

Service desk for citizens and the DPO toolset for privacy obligations.

SCOPE

- Tickets: identity verification, grievances, general enquiries; assignment, templates, SLA, escalation matrix
- Data subject requests (access, rectification, account closure) with due dates
- Retention schedule by data class, breach register with notification clock, DPIA records, policy acknowledgements

USED BY PORTALS

Public website Borrower self-service
Government — CMS & Administration
Government — Supervision

INTEGRATES WITH — INTERNAL

E02 Identity & Access Management (IAM)
E12 Dispute & Correction E16 Notification Service
E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

Email / hotline system (option)

KEY APIS

/v1/tickets
/v1/grievances
/v1/dsr
/v1/breaches

DATA OWNED

Tickets, DSRs, breaches, DPIAs

E15 Metering, Billing & Payments

Production C9, ADM-10, MFI-18, AC08

Charges for reports and subscriptions, keeps the ledger and reconciles every payment.

SCOPE

- Billable events: Basic USD 2 / Full USD 4 per borrower per institution (shared for 30 days), no charge for no-hit or retries
- Plans: pay-per-report (prepaid wallet or monthly invoice) and Unlimited-checks subscription
- Citizen fees for extra personal reports
- Checkout, wallet top-ups, receipts; monthly invoice run, credit notes, reconciliation, payment status

USED BY PORTALS

Borrower self-service MFI Member Portal

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E07 Inquiry & Credit Report Engine

E16 Notification Service

E17 Workflow & Maker-Checker

E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

Payment gateways: KBZPay, Wave Money, AYA Pay, CB Pay, MPU/Visa/Mastercard, bank transfer

Accounting / ERP (option)

KEY APIS

/v1/billing/events

/v1/checkout

/v1/wallets

/v1/invoices

/v1/subscriptions

Payment webhooks

DATA OWNED

Ledger, entitlements, wallets, invoices, payments

E16 Notification Service

Production C10, ADM-11

Every SMS, email, in-app message and webhook, from EN/MM templates with a delivery log.

SCOPE

- Channels: SMS, email, in-app, MFI webhooks (HMAC-signed)
- Templates in English and Myanmar with variables; retries; opt-out (except mandatory/security)
- Delivery log with masked recipients

USED BY PORTALS

Borrower self-service MFI Member Portal

Government — Supervision

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

SMS gateway / telcos (MPT, Ooredoo/ATOM, Mytel)

SMTP relay (government mail policy)

MFI webhook endpoints

KEY APIS

Internal: /notify; Admin: /v1/templates, /v1/deliveries

DATA OWNED

Templates, delivery log, opt-outs

E17 Workflow & Maker-Checker

Pilot SEC-03

One approval engine for every sensitive change so the rules are enforced the same way everywhere.

SCOPE

- Pending requests with diff, maker, checker role; maker can never approve their own item
- Used by: licence changes, role/user changes, rule activation, data corrections, content publishing, config, batch sign-off, identity merges, large-loan second approval
- Approvals inbox, SLA, audit of decisions; effect applied on approval

USED BY PORTALS

MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E19 Audit & Security Logging E16 Notification Service

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/approvals
/v1/approvals/{id}/decide

DATA OWNED

Approval requests, decisions

E18 Content Management (Headless CMS)

Pilot C8, CMS-01..16, ADM-02, AC07/AC11

All public and portal content, bilingual, with workflow and classification enforced by the API.

SCOPE

- Content types: pages, services, announcements, news, FAQ, publications, banners, forms, glossary
- EN/MM fields, Zawgyi detection, translation status; UI string dictionary for all portals
- Workflow draft → review → approve → schedule → publish → archive; version history and restore
- Classification (Public / MFI-only / Regulator-only / Internal) enforced server-side
- Media library with virus scan and alt text; navigation/landing layout; targeted notices with read receipts

USED BY PORTALS

Public website Borrower self-service
MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E17 Workflow & Maker-Checker
E21 Document & File Service E22 Search Service
E16 Notification Service E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

CDN for the public website

KEY APIS

/v1/content
/v1/content/{id}/publish
/v1/media
/v1/i18n/{lang}

DATA OWNED

Content items & versions, media, UI string dictionary

E19 Audit & Security Logging

Pilot C11, SEC-08, ADM-12, AC04

Tamper-evident record of who did what, to which record, for what purpose.

SCOPE

- Hash-chained audit entries (actor, role, tenant, IP, action, target, purpose, before/after, outcome) on WORM storage
- Security events: failed logins, MFA failures, IP anomalies, privileged actions, break-glass use, cross-tenant denials
- Daily integrity check; search, saved queries, CSV/signed-PDF extracts; SIEM export (syslog/JSON)

USED BY PORTALS

MFI Member Portal Government — Supervision
Government — CMS & Administration
Borrower self-service

INTEGRATES WITH — INTERNAL

None (append-only sink)

INTEGRATES WITH — EXTERNAL

SIEM / SOC (CBM or national CERT)
WORM archive storage

KEY APIS

Internal: /audit (append-only); Admin: /v1/audit, /v1/audit/verify

DATA OWNED

Audit log (10-year WORM), security log, admin change log

E20 Reporting, Analytics & Geo (DWH)

Pilot C12, GOV-01/02/14/16/17, PUB-08

One source of numbers — KPI dictionary, dashboards, statutory reports, open data and maps.

SCOPE

- Pseudonymised data warehouse fed from the registry and operational engines
- KPI dictionary with definitions, lineage and as-of dates; executive, supervisor, operations dashboards
- Lending activity, citizen adoption, report-request and billing metrics
- Township/region aggregates and boundaries for heat-maps; census layers
- Scheduled statutory reports (bulletins, scorecards), ad-hoc builder on anonymised data, public statistics and CSV downloads

USED BY PORTALS

Public website MFI Member Portal
Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E04 Borrower & Loan Registry
E05 Ingestion & Data Quality
E07 Inquiry & Credit Report Engine
E10 Loan Application Hub E12 Dispute & Correction
E15 Metering, Billing & Payments

INTEGRATES WITH — EXTERNAL

GIS boundaries (geoBoundaries / national MIMU data)
Census data sources

KEY APIS

/v1/kpis
/v1/stats (public)
/v1/reports/scheduled
/v1/geo/*

DATA OWNED

DWH (pseudonymised), aggregates, report outputs

E21 Document & File Service

Pilot CMS-08, BOR-08, MFI-06

Storage and processing for every uploaded or generated file.

SCOPE

- Object storage for batch files, evidence, consent forms, media, report PDFs
- Virus scanning, type/size limits, signed URLs, retention tags
- PDF generation (reports, receipts, invoices, statutory reports) with QR and watermark

USED BY PORTALS

Public website Borrower self-service
MFI Member Portal Government — Supervision
Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

Object storage (in-country or CBM-approved hosting)
Antivirus engine

KEY APIS

Internal: /files, /pdf/render

DATA OWNED

Files and metadata

E22 Search Service

Production PUB-13

Site-wide search for the public website and quick finders in the portals.

SCOPE

- Index of services, notices, FAQ, publications, MFI directory (public classification only)
- Myanmar-aware tokenisation; "no result" analytics for the CMS

USED BY PORTALS

Public website Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E18 Content Management (Headless CMS)
E03 Institution Master & Licensing

INTEGRATES WITH — EXTERNAL

None

KEY APIS

/v1/search

DATA OWNED

Search index

E23 Operations: Scheduler, Config & Backup

Pilot ADM-13, SEC-10, AC09

Runs the platform: jobs, configuration, feature flags, backups and disaster recovery.

SCOPE

- Job scheduler (ingestion windows, EWS runs, report generation, retention purge, invoice run)
- Configuration with change approval, feature flags, maintenance mode, API client registry
- Backups (RPO 1 h), quarterly restore tests (RTO 4 h), DR site; health monitoring and uptime

USED BY PORTALS

Government — CMS & Administration

INTEGRATES WITH — INTERNAL

E17 Workflow & Maker-Checker

E19 Audit & Security Logging

E16 Notification Service

INTEGRATES WITH — EXTERNAL

Monitoring / APM

Backup & DR infrastructure

KEY APIS

Admin: /v1/jobs, /v1/config, /v1/flags, /v1/backups

DATA OWNED

Job runs, config versions, backup records

E24 External Integration Hub

Option §11.2, G19

Adapters to national and partner systems, each behind its own agreement.

SCOPE

- NRC registry lookup for identity verification
- Data exchange with MCIX and Myanmar Credit Bureau (MMCB) to avoid double reporting
- Alternative data (telco / utility) with borrower consent
- Retry, circuit breaking and audit for every outbound call

USED BY PORTALS

Borrower self-service

MFI Member Portal

Government — Supervision

INTEGRATES WITH — INTERNAL

E06 Identity Resolution

E07 Inquiry & Credit Report Engine

E09 Consent Management

E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

National NRC registry

MCIX (ThitsaWorks / MMA)

Myanmar Credit Bureau (MMCB)

Telco / utility data providers

KEY APIS

Internal adapters only

DATA OWNED

Integration logs, mapping tables

Assistive AI kept separate from decisions that affect citizens.

SCOPE

- Public assistant answering from published content only (no personal data)
- Anomaly and over-indebtedness insights for supervisors (explainable, human-reviewed)

USED BY PORTALS

Public website Government — Supervision

INTEGRATES WITH — INTERNAL

E18 Content Management (Headless CMS)
E20 Reporting, Analytics & Geo (DWH)
E19 Audit & Security Logging

INTEGRATES WITH — EXTERNAL

LLM / ML hosting (per data-residency decision)

KEY APIS

/v1/assistant
Internal: /insights

DATA OWNED

Prompt/response logs (no PII), model versions

6. External integrations

External system	Used for	Engines	Phase
MFI core banking & loan systems	Batch submission (API/SFTP), inquiries, application webhooks	E01, E05, E07, E10	Pilot
Government SSO / directory (CBM, CIC)	Workforce sign-in and MFA	E02	Pilot
SMS gateway / telcos	OTP, notifications	E02, E16	Pilot
SMTP relay	Email notifications	E16	Pilot
eKYC / face-match provider	Citizen identity verification	E02	Production
Payment gateways (KBZPay, Wave Money, AYA Pay, CB Pay, cards, bank)	Checkout, top-ups, invoice payment	E15	Production
SIEM / SOC	Security monitoring	E19	Pilot
Object storage & antivirus	Files, evidence, PDFs	E21	Pilot
CDN / DNS / WAF provider	Public website delivery, protection	E01, E18	Pilot
GIS boundaries & census sources	Maps and township aggregates	E20	Production
National NRC registry	Authoritative identity check	E02, E06, E24	Option
MCIX / Myanmar Credit Bureau	Cross-bureau data exchange	E24	Option
Telco / utility data providers	Alternative data with consent	E24	Option
LLM / ML hosting	Assistant and insights	E25	Option

7. How the engines work together

Citizen gets a credit report

1. PUB/BOR → E02 register, verify identity (eKYC or SMS), confirmation by E16
2. BOR → E11 request report → E04 + E06 automated validation
3. ADM officer → E11 approve (E17 rules) → E07 issue snapshot → E21 PDF + QR
4. E16 SMS/email "report ready" → BOR views report; E19 logs every step
5. Anyone → PUB → E07 verify report ID + code

Citizen applies for a loan

1. BOR → E10 application + E09 digital consent
2. MFI → E10 inbox → E07 inquiry (E15 charges Basic/Full or subscription; E08 grade)
3. MFI → E10 decision (second approval via E17 above threshold) → disburse
4. E10 → E04 posts the loan; E16 notifies the citizen; E20 updates lending statistics

Monthly data submission

1. MFI system → E01 → E05 batch (idempotent) → validation report
2. MFI checker → E17 sign-off with attestation
3. E05 → E06 identity resolution → E04 load + reconciliation
4. E08 EWS evaluation → E13 alerts; E20 KPIs; E19 audit

Dispute and correction

1. BOR (or E14 helpdesk) → E12 dispute + evidence (E21 virus scan)
2. MFI → E12 response / correction within SLA
3. ADM steward → E12 approve (E17) → E04 new record version
4. E16 notifies the citizen; SLA breach → E13 consumer-protection escalation

Publishing a public notice

1. ADM editor → E18 draft (EN/MM) → E17 review by publisher
2. E18 publish (classification enforced) → CDN / E22 search index
3. Regulator notices: GOV → E18 → publication approval → PUB

8. Build phasing

Pilot 16 engines

E01 API Gateway & Edge · E02 Identity & Access Management (IAM) · E03 Institution Master & Licensing · E04 Borrower & Loan Registry · E05 Ingestion & Data Quality · E06 Identity Resolution · E07 Inquiry & Credit Report Engine · E08 Scoring & Rules Engine · E09 Consent Management · E12 Dispute & Correction · E17 Workflow & Maker-Checker · E18 Content Management (Headless CMS) · E19 Audit & Security Logging · E20 Reporting, Analytics & Geo (DWH) · E21 Document & File Service · E23 Operations: Scheduler, Config & Backup

Production 7 engines

E10 Loan Application Hub · E11 Citizen Report Request Workflow · E13 Supervision & Case Management · E14 Helpdesk, Grievance & Data Protection · E15 Metering, Billing & Payments · E16 Notification Service · E22 Search Service

Option 2 engines

E24 External Integration Hub · E25 AI Services

OPEN DECISIONS THAT AFFECT THE BACKEND

- Hosting location and data residency (in-country or CBM-approved cloud) — affects E21, E20, E25.
- Identity verification route for citizens (NRC registry API, MFI phone OTP, branch activation) — affects E02, E24.
- Payment providers and settlement accounts — affects E15.
- CIC's role versus MCIX and MNCB, and whether data exchange is in scope — affects E24.
- Exact CBM regulation clauses for retention, dispute SLA and consent — affects E04, E09, E12, E14, E19.